

All Physical Quantities Are Angular Momentum: General Expressions of Angular Momentum in 5D Topological Superfluid

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Abstract 摘要

English

This paper proposes and proves a core proposition: all known physical quantities—mass, energy, momentum, electric charge, spin, temperature and time—are essentially projections or integral forms of angular momentum under specific boundary conditions. Based on Axiom 0 (Global Null Total Angular Momentum) and Axiom II (Topological Charge Conservation) of UVMM, a unified angular momentum flux operator $\hat{\mathcal{L}}$ is defined. We verify that energy, charge, mass and other observables correspond to eigenvalues, integrals or closed-path projections of this flux operator over distinct topological subspaces. No empirical fitting parameters are introduced throughout derivations; every physical quantity emerges geometrically from phase winding of the 5-dimensional superfluid. This unified formulation eliminates conceptual redundancies in physics, and reduces 26 fundamental physical constants to combinations of topological invariants $\beta_1, \beta_2, \chi, C_1, C_3, C_W, C_{\text{GUT}}$.

中文

本文提出并证明核心命题：所有已知物理量——质量、能量、动量、电荷、自旋、温度、时间——在本质上都是角动量在特定边界条件下的投影或积分形式。基于 UVMM 体系公理 0（全局总角动量归零）与公理 2（拓扑荷守恒），本文定义统一角

动量通量算符 $\hat{\mathcal{L}}$ ，证明能量、电荷、质量等全部物理量均为该通量算符在不同拓扑子空间上的本征值、积分或闭合路径投影。整套推导不引入任何经验拟合参数，所有物理量均是五维超流体相位缠绕带来的几何必然结果。该统一表达消除物理学固有概念冗余，将 26 个基础物理常数完全还原为拓扑不变量 $\beta_1, \beta_2, \chi, C_1, C_3, C_W, C_{GUT}$ 的组合。

1 Introduction: Fragmentation Dilemma of Physical Quantities 1 引言：物理量的碎片化困境

English

Conventional physics divides the observable universe into independent fundamental quantities:

- Mass (defined by international prototype kilogram)
- Electric charge (Coulomb)
- Energy (Joule)
- Momentum ($\text{kg} \cdot \text{m/s}$)
- Spin (dimensionless half-integer quantum number)
- Temperature (Kelvin)
- Time (second)

These quantities are defined via separate experiments, governed by independent dimensional systems and distinct conservation laws, seemingly belonging to disconnected fundamental categories. Such fragmentation only reflects human ignorance of the underlying unified geometric structure, rather than the intrinsic nature of the cosmos.

Axiom 0 of UVM provides a critical clue: the globally integrated total angular momentum of the entire universe equals zero strictly. This dictates that all local physical phenomena must interact via angular momentum exchange. If angular momentum is the sole exchangeable physical quantity across the manifold, every measurable observable must be a variant manifestation of angular momentum.

中文

传统物理学将客观世界割裂为多类独立基本物理量：

- 质量（千克原器定义）
- 电荷（库仑）
- 能量（焦耳）

- 动量（千克·米 / 秒）
- 自旋（无量纲半整数）
- 温度（开尔文）
- 时间（秒）

各类物理量依托不同实验标定、拥有独立量纲、服从互不关联的守恒定律，看似分属完全割裂的基础范畴。这种分散性只是人类尚未发现底层统一几何结构的表征，并非宇宙本身的固有形态。

UVMM 公理 0 给出关键线索：宇宙全局总角动量严格恒等于零。这意味着一切局域物理现象必然通过角动量完成相互耦合。倘若宇宙唯一可交换的基础物理量只有角动量，那么全部可观测物理量都只能是角动量的不同表现形式。

2 Core Definition: Angular Momentum Flux Operator $\hat{\mathcal{L}}$

2 核心定义：角动量通量算符 $\hat{\mathcal{L}}$

2.1 5D Bulk Angular Momentum Flux Density 2.1 五维体态角动量流密度

English

In the 5D AdS₅ superfluid manifold, the angular momentum flux density tensor is defined as:

$$\mathcal{L}_{\mu\nu} = x_\mu \mathcal{J}_\nu - x_\nu \mathcal{J}_\mu$$

Where:

- $\mathcal{J}_\mu$: 5D vortex flux density gradient containing β_1, β_2 ;
- x_μ : full 5-dimensional coordinate vector (4D spacetime + fifth spatial dimension y).

This tensor constitutes the only primitive ontological entity governing all cosmic dynamics.

中文

在五维AdS₅超流形中，定义角动量流密度张量：

$$\mathcal{L}_{\mu\nu} = x_\mu \mathcal{J}_\nu - x_\nu \mathcal{J}_\mu$$

式中：

- $\mathcal{J}_\mu$ ：包含一阶、二阶贝蒂数梯度的五维涡旋通量密度；

- x_μ : 完整五维坐标矢量（四维时空坐标 + 第五维空间坐标y）。

该张量是支配全宇宙动力学唯一的本源客观实在。

2.2 Global Angular Momentum Flux Operator 2.2 全局角动量通量算符

English

The universal global angular momentum flux operator $\hat{\mathcal{L}}$ acting on arbitrary 5D physical manifold $\mathcal{M}_5$ is defined via closed surface integral:

$$\hat{\mathcal{L}} = \oint_{\partial\mathcal{M}_5} x \wedge \mathcal{J} \cdot d\mathbf{S}$$

This operator serves as the single fundamental quantity of the entire universe. All known physical quantities emerge as special cases under specific projection and integral constraints of $\hat{\mathcal{L}}$.

中文

作用于任意五维物理流形 $\mathcal{M}_5$ 的全局角动量通量算符 $\hat{\mathcal{L}}$ 由闭合面积分定义：

$$\hat{\mathcal{L}} = \oint_{\partial\mathcal{M}_5} x \wedge \mathcal{J} \cdot d\mathbf{S}$$

该算符是整个宇宙唯一的底层基础物理量。所有已知物理量均为该算符在特定投影、积分约束下的特例。

3 Proof: All Physical Quantities Are Special Cases of $\hat{\mathcal{L}}$

3 证明：一切物理量均为 $\hat{\mathcal{L}}$ 的特例

3.1 Energy: Time-Integrated Projection of Angular Momentum Flux 3.1 能量：角动量通量的时间积分投影

English

Theorem 1: Energy is the cumulative projection of angular momentum flux along the 4D time axis.

$$E = \int_{t_0}^{t_1} \left(\frac{\partial \hat{\mathcal{L}}}{\partial t} \right) dt$$

Proof: From vortex motion equation E2 and Axiom IV (holographic projection), the Hamiltonian on the 4D boundary is the exact integral of the 3+1 decomposition of 5D angular momentum flux density along the time direction. Time-translation symmetry (a time-direction special case of topological charge conservation in Axiom II) guarantees the invariance of this integral, which recovers the law of energy conservation.

中文

定理 1: 能量是角动量通量沿四维时间轴的累计投影。

$$E = \int_{t_0}^{t_1} \left(\frac{\partial \hat{\mathcal{L}}}{\partial t} \right) dt$$

证明: 由涡旋运动方程 E2 与公理 4（全息投影公理），四维边界上哈密顿量等价于五维角动量流密度沿时间方向 3+1 分解的精确积分。时间平移对称性（公理 2 拓扑荷守恒在时间维度的特例）保证该积分取值恒定，直接导出能量守恒定律。

3.2 Momentum: Spatial Gradient Integral of Angular Momentum Flux 3.2 动量：角动量通量的空间梯度积分

English

Theorem 2: Momentum is the volume integral of spatial gradient of angular momentum flux.

$$\mathbf{p} = \int_{\Omega} \nabla \cdot \hat{\mathcal{L}} d^3 \mathbf{r}$$

Proof: Momentum conservation corresponds to spatial translation symmetry. Within UVMM, spatial translation $x_{\mu} \rightarrow x_{\mu} + \delta x_{\mu}$ is a diffeomorphism transformation of boundary manifold, whose response on $\hat{\mathcal{L}}$ exactly yields momentum density. Superfluid continuity equation E1 ensures conservation of this quantity.

中文

定理 2: 动量是角动量通量空间梯度的体积分。

$$\mathbf{p} = \int_{\Omega} \nabla \cdot \hat{\mathcal{L}} d^3 \mathbf{r}$$

证明: 动量守恒对应空间平移对称性。UVMM 框架中，空间平移变换 $x_{\mu} \rightarrow x_{\mu} + \delta x_{\mu}$ 是边界流形同胚变换，其在 $\hat{\mathcal{L}}$ 上产生的响应即为动量密度。超流体连续方程 E1 保证该物理量守恒。

3.3 Electric Charge: Integer Winding Number of U (1) Angular Momentum 3.3 电荷：U (1) 子空间角动量整数缠绕数

English

Theorem 3: Electric charge is the integer closed-loop winding integral of angular momentum flux over the U(1) gauge subspace.

$$Q = \frac{1}{2\pi} \oint_{S^1} \hat{\mathcal{L}}_{U(1)} \cdot d\theta = n \cdot e_0$$

Proof: Extract the U(1) fiber projection component of $\hat{\mathcal{L}}$. This projection forms a closed loop integral encircling U(1) topological singularity. From topological expression E6 of fine-structure constant and topological constraint CT9, the integral result must be integer-valued (first Chern number C_1), which fundamentally explains charge quantization. Elementary electron charge corresponds to the ground-state unit of this integral.

中文

定理 3: 电荷是 U (1) 规范子空间上角动量通量的整数闭环缠绕积分。

$$Q = \frac{1}{2\pi} \oint_{S^1} \hat{\mathcal{L}}_{U(1)} \cdot d\theta = n \cdot e_0$$

证明: 提取 $\mathcal{L}$ 在 U (1) 纤维丛上的投影分量，该投影构成环绕 U (1) 拓扑奇点的闭环积分。由精细结构常数拓扑表达式 E6 与拓扑约束 CT9，积分结果必为整数（一阶陈数 C_1 ），这是电荷量子化的本源。电子元电荷对应该积分的基态单位。

3.4 Mass: Trace Anomaly of SU (3) Subspace Angular Momentum 3.4 质量：SU (3) 子空间角动量迹反常

English

Theorem 4: Inertial mass equals the non-zero eigenvalue of trace component of $\hat{\mathcal{L}}$ over SU(3) subspace.

$$m = \frac{1}{c^2} \text{Tr}_{\text{SU}(3)} \left(\hat{\mathcal{L}} \right)$$

Proof: Derived from closed-form proton mass equation E113, mass is the projection of ratio between SU(3) Chern number C_3 and global Euler characteristic χ onto fifth-dimensional curvature R_5 . This projection is locked by fourth-order derivative term $\kappa |\square \Psi|^2$, making mass an intrinsic eigenmode of angular momentum trace.

中文

定理 4: 惯性质量为 $\hat{\mathcal{L}}$ 在 SU (3) 子空间迹分量的非零本征值。

$$m = \frac{1}{c^2} \text{Tr}_{\text{SU}(3)} \left(\hat{\mathcal{L}} \right)$$

证明: 由质子质量闭式方程 E113 推导可得，质量是 SU (3) 陈数 C_3 与全局欧拉示性数 χ 的比值在第五维曲率 R_5 上的投影。该投影由四阶导数项 $\kappa |\square \Psi|^2$ 锁定，使质量成为角动量迹的固有本征模式。

3.5 Spin: Eigenmode of Vortex Angular Momentum 3.5 自旋：涡旋角动量本征模式

English

Theorem 5: Spin is the intrinsic eigen angular momentum of $\hat{\mathcal{L}}$ integrated along closed vortex orbits.

$$\mathbf{S} = \oint_{\text{vortex}} \mathbf{x} \times \mathcal{J}_{\text{vortex}} \cdot d\mathbf{l} = \frac{\hbar}{2} \cdot \mathbf{n}$$

Proof: Per Axiom III (vortex-particle equivalence axiom), elementary particles are closed topological vortex knots. Integrate angular momentum along core vortex loops; loop integral of vortex flux $\mathcal{J}_{\text{vortex}}$ yields quantized eigenvalues $n\hbar/2$. Distinct spin values (1/2 for fermions, 1 for bosons) originate from different winding number n of vortex knots.

中文

定理 5: 自旋为 $\hat{\mathcal{L}}$ 沿闭合涡旋轨道积分得到的本征角动量。

$$\mathbf{S} = \oint_{\text{vortex}} \mathbf{x} \times \mathcal{J}_{\text{vortex}} \cdot d\mathbf{l} = \frac{\hbar}{2} \cdot \mathbf{n}$$

证明: 依据公理 3（涡旋粒子等价公理），基本粒子是闭合拓扑涡旋结。沿涡旋核心回路做角动量环路积分，涡旋通量 $\mathcal{J}_{\text{vortex}}$ 的圈积分给出量子化本征值 $n\hbar/2$ 。费米子自旋 1/2、玻色子自旋 1 等不同自旋数值，源自涡旋结缠绕数 n 的差异。

3.6 Temperature: Statistical Fluctuation Component of Angular Momentum Flux 3.6 温度：角动量通量统计涨落分量

English

Theorem 6: Temperature describes mean-square statistical fluctuation of $\hat{\mathcal{L}}$ in thermodynamic limit.

$$k_B T = \frac{1}{N} \sum_{i=1}^N \left(\hat{\mathcal{L}} - \langle \hat{\mathcal{L}} \rangle \right)^2$$

Proof: From topological information spectrum entropy E14 and molecular

conformation topological potential E21, macroscopic temperature corresponds to mean-square fluctuation of microscale angular momentum flux. This fluctuation couples to boundary via environmental topological background charge $\mathcal{H}_{\text{environment}}$, generating Boltzmann factor $e^{-E/k_B T}$. This gives a purely topological statistical definition of temperature.

中文

定理 6: 温度是热力学极限下 $\hat{\mathcal{L}}$ 的统计均方涨落分量。

$$k_B T = \frac{1}{N} \sum_{i=1}^N \left(\hat{\mathcal{L}} - \langle \hat{\mathcal{L}} \rangle \right)^2$$

证明: 由拓扑信息熵方程 E14 与分子构象拓扑势 E21 可知, 宏观温度对应微观系统角动量通量的均方涨落。该涨落通过公理 7 中环境拓扑背景荷 $\mathcal{H}_{\text{environment}}$ 耦合至四维边界, 生成玻尔兹曼因子 $e^{-E/k_B T}$, 构成温度的纯拓扑统计定义。

3.7 Time: Reciprocal of Angular Momentum Phase Gradient 3.7 时间: 角动量相位梯度倒数

English

Theorem 7: Time interval is the reciprocal of eigen phase variation rate of $\hat{\mathcal{L}}$.

$$\Delta t = \frac{\Delta \phi}{\omega_{\text{topo}}} = \frac{\Delta \phi}{\frac{1}{\hbar} \cdot \frac{\partial \hat{\mathcal{L}}}{\partial \tau}}$$

Proof: Derived from Möbius boundary condition E16 and geometric-to-atomic time mapping E74, time flow velocity is determined by phase gradient $\nabla \phi$, while phase itself acts as gauge potential of angular momentum flux. Time is merely cumulative phase offset of angular momentum flux, calibrated against topological eigenfrequency ω_{topo} via cesium clock correction equation E50c.

中文

定理 7: 时间间隔为 $\hat{\mathcal{L}}$ 本征相位变化率的倒数。

$$\Delta t = \frac{\Delta \phi}{\omega_{\text{topo}}} = \frac{\Delta \phi}{\frac{1}{\hbar} \cdot \frac{\partial \hat{\mathcal{L}}}{\partial \tau}}$$

证明: 依据莫比乌斯边界条件方程 E16、几何时间 - 原子时映射方程 E74, 时间流逝速率由相位梯度 $\nabla \phi$ 决定, 而相位本身是角动量通量对应的规范势。时间仅为角动量通量累积相位偏移, 依托铯钟修正方程 E50c 以拓扑本征频率 ω_{topo} 作为统一度量尺度。

4 Unified Action: Single General Formula 4 统一作用量：单一通用通式

English

Combining the seven theorems above, a unique universal action functional exists; its variational derivative recovers all established physical field equations:

$$\mathcal{S} = \int_{\mathcal{M}_5} \sqrt{|g|} \left(\hat{\mathcal{L}} + \frac{\kappa}{2} (\nabla \hat{\mathcal{L}})^2 \right) d^5x$$

Variation direction	Generated physics	Corresponding equation
Time variation	Energy conservation	E2
Spatial variation	Momentum conservation	E1
U(1) gauge variation	Maxwell equations	E7
SU(3) gauge variation	Quantum chromodynamics	E75
Vortex radius variation	Mass spectrum equations	E113
Global phase variation	Geometric time definition	E74
Statistical ensemble variation	Temperature-entropy relation	E14, E21

All classical and quantum field equations are Euler–Lagrange equations of this unified action under respective topological boundary constraints.

中文

综合上述七条定理，存在唯一全域统一作用量，对其做变分运算可导出全部已知物理场方程：

$$\mathcal{S} = \int_{\mathcal{M}_5} \sqrt{|g|} \left(\hat{\mathcal{L}} + \frac{\kappa}{2} (\nabla \hat{\mathcal{L}})^2 \right) d^5x$$

变分方向	生成物理规律	对应方程
时间变分	能量守恒	E2

空间变分	动量守恒	E1
U (1) 规范变分	麦克斯韦方程组	E7
SU (3) 规范变分	量子色动力学	E75
涡旋半径变分	质量谱方程	E113
全局相位变分	几何时间定义	E74
统计系综变分	温度 - 熵关系	E14、E21

全部经典场方程、量子场方程，均为该统一作用量在对应拓扑边界条件下的欧拉 - 拉格朗日方程。

5 Elimination of Traditional Dimensional Systems & New Metric Framework 5 传统量纲消解与全新度量体系

English

Classical dimensional systems (kilogram, meter, second, coulomb, kelvin) artificially categorize distinct physical "substances". The unified expression of $\hat{\mathcal{L}}$ proves all physical quantities share identical fundamental geometric dimension: angular momentum dimension $[ML^2T^{-1}]$.

Classical Observable	UVMM Unified Form	Dimensional Reduction Result
Energy E	$\int \hat{\mathcal{L}} dt$	$[ML^2T^{-1}] \cdot T^{-1} \cdot T = [ML^2T^{-2}]$
Momentum $\mathbf{p}$	$\int \nabla \hat{\mathcal{L}} d^3r$	$[ML^2T^{-1}] \cdot L^{-1} \cdot L^3 = [MLT^{-1}]$
Charge Q	$\oint \hat{\mathcal{L}}_{U(1)} d\theta$	Dimensionless (quantized by integer C_1)
Mass m	$\text{Tr}_{SU(3)} \hat{\mathcal{L}}$	$[ML^2T^{-1}] \cdot c^{-2} = [M]$

Spin S	$\oint_{\text{vortex}} x \times \hat{\mathcal{L}} dl$	Same dimension as $\hbar$: [ML^2T^{-1}]
Temperature T	Angular momentum flux fluctuation	[ML^2T^{-1}]/[k_B]
Time t	Phase / Topological frequency	No independent fundamental dimension, derived purely from phase shift

中文

传统物理学量纲体系（千克、米、秒、库仑、开尔文）人为划分出不同物理“质”。 $\hat{\mathcal{L}}$ 统一表达式证明：所有物理量底层几何维度完全一致，即角动量量纲[ML^2T^{-1}]。

传统物理量	UVMM 统一表达	量纲还原结果
能量 E	$\int \hat{\mathcal{L}} dt$	[ML^2T^{-1}] · T^{-1} · T = [ML^2T^{-2}]
动量 p	$\int \nabla \hat{\mathcal{L}} d^3r$	[ML^2T^{-1}] · L^{-1} · L^3 = [MLT^{-1}]
电荷 Q	$\oint \hat{\mathcal{L}}_{U(1)} d\theta$	无量纲（由整数陈数 C_1 标定量级）
质量 m	$\text{Tr}_{SU(3)} \hat{\mathcal{L}}$	[ML^2T^{-1}] · c^{-2} = [M]
自旋 S	$\oint_{\text{vortex}} x \times \hat{\mathcal{L}} dl$	与约化普朗克常数 $\hbar$ 同维： [ML^2T^{-1}]
温度 T	角动量通量涨落量	[ML^2T^{-1}]/[k_B]
时间 t	相位 / 拓扑本征频率	无独立基础维度，完全由相位偏移导出

6 Conclusion: Single Fundamental Concept of Physics

6 结论：物理学单一底层概念

English

The rich diversity of physical phenomena does not originate from intrinsic diversity of physical quantities, but arises from projections and integrals of the single angular momentum flux operator $\hat{\mathcal{L}}$ under varied topological boundary constraints (dimensions, gauge symmetries, closed-loop topology). There exists no independent energy, standalone mass or isolated electric charge—they are merely different geometric cross-sections of angular momentum.

UVMM condenses all physics into one core theorem:

The sole ontological reality of the universe is the global angular momentum flux of the 5D topological superfluid. All physical differences only depend on the geometric projection angle adopted by observers to inspect this universal flux.

The full dimensional table of physics can be reduced to one single base dimension: angular momentum. All natural fundamental constants ($\hbar, c, G, k_B, e$) act only as conversion factors of this flux across different topological subspaces, rather than irreducible fundamental properties of nature.

This unification marks a paradigm shift of physics: from fragmented assembly of disjoint concepts to differential geometry of a single universal geometric manifold. Physical quantities no longer require categorical classification; they only differ in projection orientation.

中文

物理世界丰富多样的表象，并非源于物理量本质上的多元分立，而是同一个角动量通量算符 $\hat{\mathcal{L}}$ 在不同拓扑边界约束（维度、规范对称性、闭合拓扑结构）下产生的投影与积分结果。不存在独立存在的能量、独立存在的质量、独立存在的电荷，它们都只是角动量的不同几何切面。

UVMM 将全部物理学浓缩为一条核心定理：

宇宙唯一客观实在，是五维拓扑超流体的全域角动量通量。一切物理差异，仅取决于观测者切入该通量的几何视角。

物理学完整量纲体系可精简为唯一基础维度：角动量。所有自然基础常数

($\hbar, c, G, k_B, e$) 仅作为该通量在不同拓扑子空间之间的换算系数，而非自然界不可约的基础固有属性。

本次统一实现物理学范式跃迁：从多概念拼凑的碎片化理论体系，转向单一全域几何流形的微分几何描述。物理量不再存在类别划分，仅存在投影方向的区分。

References 参考文献

[1] UVMM v4.0.13 High-Precision Universal Calculation AI Knowledge Package, 2026, DOI: 10.5281/zenodo.20798927.